

Divergence-form and wide-halo mEVP in the GPU version of FESOM2: what was built, and what it costs

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1. What this note is about

Two of your proposals were implemented in our GPU version of FESOM2 and measured: **(a)** the **reformulation of the mEVP sub-cycling** in which the quantity carried from one sub-cycle to the next is the divergence of the stress, held at nodes, instead of the three stress components held at elements; and **(b)** the **wide halo**, in which the halo is exchanged only every K -th sub-cycle and the missing information is recomputed locally in a ring of ghost cells K deep, so that the result is identical to exchanging after every sub-cycle.

A third option came out of the measurements and had to be included: simply exchanging every K -th sub-cycle and letting the halo go stale. It is not exact, but it is by far the cheapest, so its accuracy had to be established rather than assumed.

Everything below is measured on production configurations. Where something is not measured, that is said.

2. Terms, configurations, meshes, machine

The sub-cycle and the halo. mEVP advances the ice momentum with 120 sub-cycles per model time step in all our configurations. The mesh is divided among MPI processes; each process owns a piece of it and keeps a one-cell-wide copy (the *halo*) of its neighbours' edge values. In the standard scheme the ice-velocity halo is exchanged after *every* sub-cycle, so one model step costs 120 halo exchanges.

The five configurations. These names are used throughout.

name	carried between sub-cycles	halo exchange
standard	stress $\sigma_{11}, \sigma_{12}, \sigma_{22}$ at elements	after every sub-cycle
divergence form	$\nabla \cdot \sigma$ (two components) at nodes	after every sub-cycle
wide halo, standard form	stress at elements, ring K deep	every K -th sub-cycle, <i>exact</i>
wide halo, divergence form	$\nabla \cdot \sigma$ at nodes, ring K deep	every K -th sub-cycle, <i>exact</i>
delayed exchange	stress at elements	every K -th sub-cycle, <i>stale halo</i>

The two wide-halo configurations are exact in the strict sense: they reproduce the every-sub-cycle trajectory bit for bit. The delayed exchange does not, and section 6.5 says by how much.

Meshes and machine. All four are production configurations. Levante at DKRZ: GPU nodes carry four NVIDIA A100 (80 GB) and one MPI process per GPU; CPU nodes carry two AMD EPYC 7763, 128 cores each. Every comparison is made inside one job allocation with the same executable, and is the faster of two repetitions; repetition spread is below 0.4% on GPU and about 1.5% on CPU.

	surface nodes	elements	levels	time step
CORE2	126 858	251 746	48	1800 s
fArc	638 387	1 270 173	48	900 s
DARS	3 160 340	6 294 801	57	120 s
NG5	7 402 886	14 827 217	70	180 s

3. The problem

One model step of the standard scheme performs 120 ice-velocity halo exchanges, each moving very little data, with very little arithmetic between them. Koldunov et al. (2019, GMD 12, 3991) diagnosed this on the fArc mesh and proposed — without testing it — “wider halo regions to reduce communication frequency”. The wide halo above is that proposal.

We can now put numbers on the diagnosis in our own code.

On CPU, at 2304 cores on fArc, the ice part is 13.2% of the model step, and inside the ice dynamics the time spent *waiting* for MPI is 4.8 ms per step against 3.5 ms of actual computation. The cost is idle time, not arithmetic.

On GPU the same structure costs more, for a second reason. Each exchange is a sequence of small GPU operations — gather the values to be sent, copy, scatter the values received — and each carries a fixed issue cost independent of how little data it touches. On CORE2 with 8 GPUs the ice-dynamics phase takes 8.2 ms per model step, of which only about 1.25 ms is arithmetic, spread over roughly 480 separate GPU operations.

So the mEVP sub-cycle, as it stands, is limited by communication and by the number of operations issued — not by arithmetic and not by memory traffic. This one fact explains every result below, including the negative ones.

4. The code before this work

FESOM2 (ocean and ice) has been rewritten in C++ with Kokkos and runs entirely on the GPU, with no per-step transfer of prognostic fields to the host. On CPU it is bit-for-bit identical to our validated C port of FESOM2, which was itself validated against Fortran FESOM2; on GPU it differs only at the level of summation order, checked against the CPU version over multi-decadal integrations. mEVP was already ported and validated, exchanging after every sub-cycle and keeping the stress at elements — that is, it was the “standard” row above, and it is the reference for everything that follows.

5. What was implemented

All four new configurations are selected at run time and are off by default, so the validated path is untouched when they are not requested.

The divergence form. The stress recursion is replaced by a recursion for $R = \nabla \cdot \sigma$ at nodes, with the ice-element mask applied so that R is advanced only where ice elements contribute.

The wide halo, both forms. At the start of each model step one extended exchange fills a ring of ghost nodes and ghost elements K cells deep with the owner’s values of the eleven fields the sub-cycle needs. Over the next $K - 1$ sub-cycles each process advances that ring itself, in the owner’s order, and needs no communication; on the K -th sub-cycle the ring is refreshed. The ring’s geometry, boundary mask and per-step element quantities are shipped from their owners rather than recomputed — a locally recomputed value can differ in the last bit, and that is enough to destroy exactness.

The delayed exchange. One conditional: the exchange fires only when the sub-cycle index is a multiple of K . Nothing else changes.

How correctness was established. For the exact options the requirement is strict: with the option on, the model must produce *bit-for-bit* the same output as with it off. That was verified for the wide halo in both forms at $K = 4$ and $K = 8$, and separately for the same machinery under standard EVP. The options were also shown to leave the model unchanged when a different vertical

mixing scheme or vertical coordinate is selected, and the GPU version was checked against the CPU version with the options on. The delayed exchange is an approximation, so bit-for-bit equality is not available and its accuracy was measured instead (section 6.5).

6. Results

6.1 The reformulation is exact in the discretisation

Carrying R and σ simultaneously and comparing them sub-cycle by sub-cycle, R agrees with $\nabla \cdot \sigma$ to $1\text{--}2.5 \times 10^{-15}$ relative, and this does not grow over 20 model steps, i.e. 2400 sub-cycles. The algebra survives the discretisation to round-off, not merely to plotting accuracy.

One genuine non-equivalence is worth naming. Within a model step the ice-element mask is fixed and the two forms are equivalent. *Across* steps, when an element enters or leaves the ice mask, they are not exactly equivalent, because in the divergence form the recursion has already been applied after summation over the elements around a node rather than before. We measured it: it fires only on steps where mask membership actually changes, and is bounded by 2.1×10^{-11} relative. That is far below physical significance, but it means the divergence form is not bitwise reproducible against the standard form over a long integration, and should be introduced as a scheme change rather than as an optimisation.

6.2 The reformulation alone does not save time

On CPU the difference is within noise everywhere we measured: between -0.34% and $+0.07\%$ of the model step, from 1 to 512 processes. On GPU it is a small *cost*: the ice-dynamics phase grows by 13% on CORE2 with 8 GPUs and by about 7% on NG5 — though, that phase being a minority of the step, only $+0.02$ to $+1.6\%$ of the step itself.

The reason is section 3. To bound the largest saving the reformulation could possibly deliver we ran a control: three element-sized arrays that the mEVP path writes every sub-cycle and never reads were simply not written. That removes strictly more memory traffic than the reformulation does, and it changed the run time by **exactly 0.00%**. There was no memory-traffic saving available to collect. Profiling agrees from the other side: total GPU arithmetic time is identical in the two forms, 21.8 against 21.7 ms per step.

This is a clean negative result rather than a disappointing one. It says the sub-cycle’s cost is not where the reformulation acts.

6.3 The wide halo works, and is worth a few percent on GPU

Figure 1 is the summary of the speed results. Every mesh there is shown at *its own operating point* — the largest node count at which the model still gains from more GPUs, read off the measured strong-scaling curve. That matters: at a common large node count these numbers roughly double, but a configuration past its scaling limit is not one anybody runs, and quoting it would flatter every option in this study.

Reading the figure:

- The wide halo is a real gain and it is *exact* — there is no accuracy argument to make — but at realistic operating points it is worth -1.8 to -6.8% of the model step, not the double-digit numbers reachable past the scaling limit.
- It is worth substantially less than the delayed exchange on every mesh.
- On CPU it never wins, on any mesh, at any K .

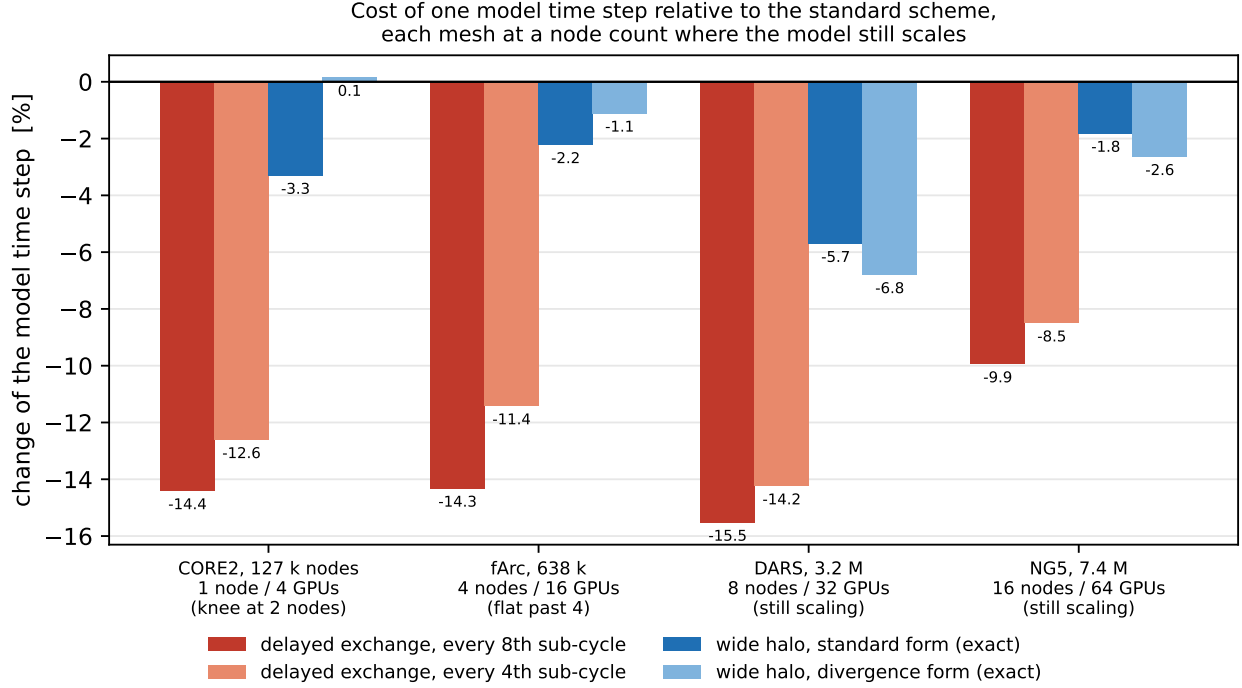


Figure 1: Cost of one model time step relative to the standard scheme (exchange after every sub-cycle). Negative is faster. **Each mesh is shown at its own operating point** — the largest node count at which the model still gains from more GPUs, read off the measured strong-scaling curve — rather than at a common node count, which would have flattered every option: CORE2 on 16 nodes gives -25.5% for the delayed exchange and is 45% slower there than on two nodes. Each mesh was measured in a single allocation with all configurations run back to back. The two wide-halo bars are *exact* — they reproduce the standard trajectory bit for bit — while the two delayed bars are approximations whose accuracy is Figure 3.

- It needs large K . At $K = 2$ it is a net *cost* on most configurations, because the duplicated ring has not been amortised. This is the opposite of what the delayed exchange wants (section 6.5), so the two options do not share a best K .

This reproduces, and sharpens, an earlier assessment. Our optimisation campaign measured the wide halo under standard EVP a month ago — NG5, 4 and 16 GPU nodes — and got $+1.7\%$ at $K = 4$ and -2.2% at $K = 8$, i.e. “low yield”. Re-measured this week at the same point with the current code: $+1.69\%$ and -2.26% . The earlier number was right. What that campaign also noted at the time, and never acted on, was that its wide refresh sent *two messages per neighbour* and that fusing them into one was a candidate fix; doing that now takes NG5 from -2.26% to -4.19% at $K = 8$ and from $+1.69\%$ to -2.47% at $K = 4$. So the fusion is worth roughly 2 to 4 percentage points — real, and not enough to change the verdict on NG5.

The CPU result has a clear mechanism, and it is the most useful thing the wide halo taught us. Trading communication for local recomputation is profitable only while the ring stays small relative to the sub-domain *and* a message stays expensive relative to the recomputation. The controlling quantity is the ratio of ghost cells that must be advanced to owned cells, which grows like $K \times (\text{perimeter}/\text{area})$ of the sub-domain. On GPU, where a process holds a large piece of the mesh, that ratio is 0.03 to 0.25 and the wide halo pays. On CPU, where processes are small and messages are comparatively cheap, it is 0.19 to 1.18 and it never did: on fArc with 2048 processes at $K = 4$ the ratio reaches 1.18 — the ring a process must advance is *larger than its own sub-domain* — and that configuration costs 5 to 7% instead of saving anything.

A second effect pushes the same way. The ring is a different size on every process, so duplicating it introduces a new load imbalance on top of the one it removes: on fArc CPU the ice-dynamics waiting time rose from 8.7 to 25.3 ms per step when the ring was made deep.

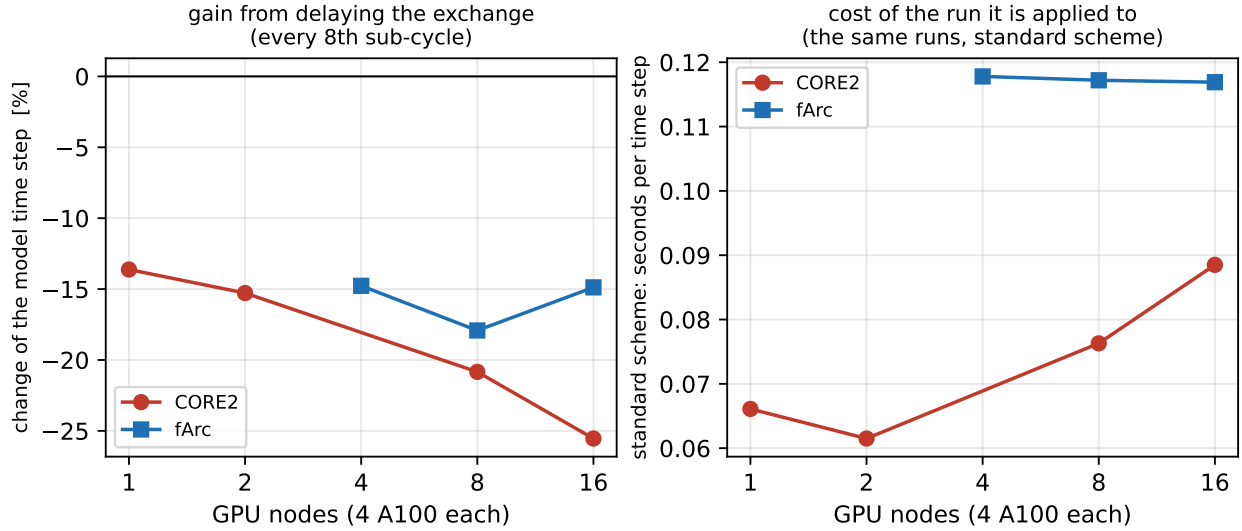


Figure 2: Why the gain from delaying the exchange varies. *Left:* change of the model time step from exchanging every 8th sub-cycle instead of every one. *Right:* the cost of the very same runs under the standard scheme. The gain is largest where the model has stopped scaling and communication dominates — CORE2 past two nodes, where adding GPUs makes the model *slower*, is exactly where the delayed exchange recovers the most.

6.4 The delayed exchange is the largest gain

Accepting a halo up to $K - 1$ sub-cycles old removes 105 of the 120 exchanges per step at $K = 8$ and costs one conditional. At the four operating points of Figure 1 it is worth -14.4 , -14.3 , -15.5 and -9.9% of the whole model step at $K = 8$ (CORE2, fArc, DARS, NG5), and -12.6 , -11.4 , -14.2 and -8.5% at $K = 4$. On CPU the same change is worth -1 to -3.4% , because a CPU process holds far less of the mesh and the ice exchange is a much smaller share of its step. For scale, the largest single code optimisation this port had previously achieved was about 8%.

$K = 4$ **retains about 90% of the gain of $K = 8$** , which matters because the accuracy evidence pushes the other way.

Figure 2 shows why the gain varies, and it is a caution as much as an explanation. It is not simply larger with more processes: it tracks how much communication the run is already paying for. CORE2 on GPU stops scaling at two nodes and then *degrades* — more GPUs, slower model — and that is exactly where the delayed exchange recovers the most, -25.5% at 16 nodes. That number is real but useless: it is measured in a configuration one would never run, and it is why Figure 1 quotes operating points instead. The same variable works in the other direction too — on NG5 spread over only four nodes, 460 000 mesh nodes per GPU, nothing we tried moved the model by more than 3%.

The mechanism was measured directly, and it is the one the 2019 paper describes: on fArc with 2304 CPU cores the ice-dynamics waiting time falls from 4.8 to 1.9 ms per step (a 60% cut) while the computing time barely moves, and MPI operations per step fall from 240 to 30. **The gain comes from removing idle time, not from computing faster.**

6.5 What the delayed exchange costs in accuracy

This decides whether the option is usable, so the test was specified in advance and then measured: one-year CORE2 integrations at $K = 2, 4, 8$ against the same run exchanging after every sub-cycle, on the CPU backend, which is exactly reproducible — so every difference seen is the approximation and nothing else. (On GPU, run-to-run differences in summation order accumulated over a year are the same size as the effect being measured, so the GPU cannot answer this question; we confirmed

it agrees to that resolution.)

The yardstick is mEVP’s own reproducibility: the difference between two independent, both correct, implementations of the same scheme — our C port and Fortran FESOM2 — over the same year.

pattern correlation after 1 year	SST	SSH	ice conc.	ice thickness	u_{ice}	v_{ice}
C port versus Fortran (the yardstick)	1.00000	1.00000	0.99999	0.99998	0.95438	0.93908
delayed exchange, $K = 2$	1.000000	1.000000	0.999996	0.999998	0.970111	0.952946
delayed exchange, $K = 4$	1.000000	1.000000	0.999995	0.999994	0.963328	0.950003
delayed exchange, $K = 8$	1.000000	1.000000	0.999992	0.999981	0.961359	0.944857

Every field stays inside the yardstick at every $K \leq 8$ (Figure 3). But the margin should be read, not just the verdict: on ice velocity the approximation sits only 1.1 to 1.5 times inside it, and at $K = 8$ ice thickness is only 1.05 times inside. We had expected an order of magnitude and were wrong. The delayed halo’s effect on the velocity field is of the same order as the spread between two implementations of mEVP — comfortably inside it at $K \leq 8$, but not by much.

Two further results:

- The error is **monotone in K** when measured as RMS difference or correlation. (It is not monotone in the maximum pointwise difference, which is carried by one to five points at the ice edge and is not a usable statistic here.)
- **Integrated ice quantities are untouched.** The largest monthly deviation in Northern Hemisphere sea-ice extent at $K = 8$ is $0.022 \times 10^6 \text{ km}^2$ against a seasonal range of 7.8 to $17.5 \times 10^6 \text{ km}^2$ — 0.23% of the amplitude. Surface temperature, salinity and sea surface height correlate at 1.000000 at every K : over a year the change does not reach the ocean surface climate.

On this evidence we would use $K = 4$: it keeps most of the speed, and since the error is monotone in K , the smallest K that pays is the defensible choice. *This is measured at CORE2 resolution only*; see section 8.

6.6 Why the divergence form’s expected advantage did not survive

This is the result we would most like your view on, because it corrects something we believed until two days ago.

The two wide-halo forms differ in what they must send. The standard form must send node velocities *and* the ghost ring’s element stresses — in our implementation, two messages per neighbour. The divergence form sends one message, because everything it carries lives at nodes, and it sends about half as many numbers. We measured the divergence form to be the faster of the two on every GPU mesh, by up to 7 percentage points of the model step, and attributed that to the smaller volume — which is the advantage the reformulation is designed to deliver.

That attribution was wrong. We changed the standard form to pack its two pieces into a *single* message per neighbour — identical values in identical order, verified bit for bit against the two-message version — and the difference reversed. Holding the volume fixed and removing one message is worth up to 8.6 percentage points of the model step; holding the message count fixed and removing 19% of the data is worth nothing. On DARS with 32 GPUs the two configurations then send *exactly* the same number of messages (24 400 per step) while the divergence form sends 19% fewer numbers — and the divergence form is 1.3 percentage points *slower*, because it must reconstruct the ghost ring’s stresses in an extra pass.

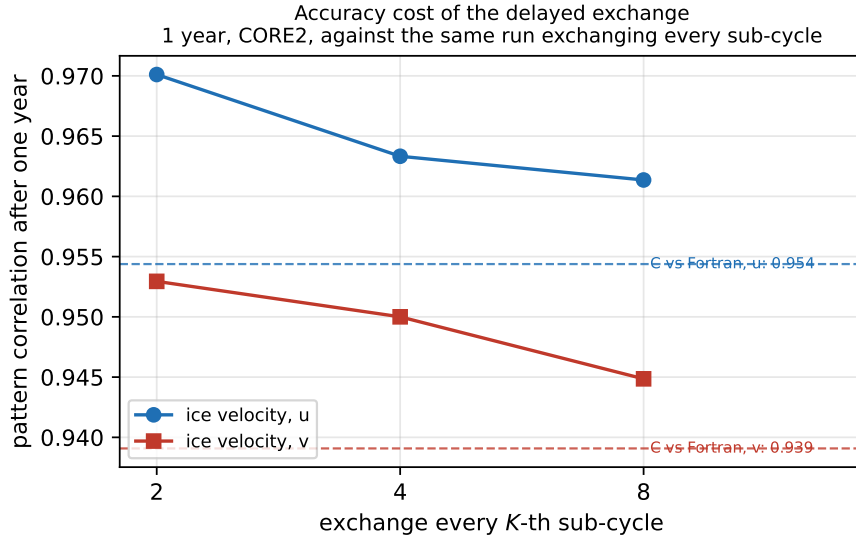


Figure 3: Accuracy cost of the delayed exchange on the field that carries it. One-year CORE2 integrations on the CPU backend (exactly reproducible, so the whole difference is the approximation), against the same run exchanging after every sub-cycle. Dashed lines: the same correlation between two independent and both-correct implementations of mEVP — our C port and Fortran FESOM2 — over the same year. The approximation stays inside that spread at every K tested, by a factor of 1.1 to 1.5.

Across ten configurations where all three variants ran side by side, removing the message helps everywhere (-0.1 to -8.6 percentage points), while halving the data is worth between -1.3 and $+2.8$ points and favours the *standard* form in eight of the ten.

The conclusion is not that the reformulation is wrong — section 6.1 shows it is exact — but that on this hardware the cost of an exchange is dominated by the per-message overhead rather than by its volume, so halving the volume buys nothing that the extra local pass does not immediately give back. Across fourteen configurations the two forms now sit within about 1.5 percentage points of each other, with the standard form ahead in ten of them; where the divergence form wins — DARS and NG5 — it wins by about one point. Neither is systematically better.

7. Where this leaves the three ideas

	status	our reading
divergence form	exact to round-off; no speed benefit on either backend	correct and elegant, but it acts on a cost the sub-cycle does not have. It would matter where memory traffic or message volume limits the sub-cycle; nothing we can run is in that regime.
wide halo (either form)	exact, bit for bit; -1.8 to -6.8% of the model step at GPU operating points, and it needs $K = 8$; never profitable on CPU	usable today where exactness is required, and the only option that needs no accuracy argument. The decision rule is the ghost-to-owned ratio, $K \times \text{perimeter/area}$ of the sub-domain, together with how expensive a message is on the machine.
delayed exchange	approximate; -8.5 to -15.5% at GPU operating points, -1 to -3.4% on CPU; error inside mEVP's own implementation spread at $K \leq 8$	the best value by a wide margin, and the one we would put into production at $K = 4$ — subject to the high-resolution check below.

8. What has not been measured

- **The accuracy of the delayed exchange at high resolution.** It is established at CORE2 resolution over one year. The speed results that motivate its use are on fArc, DARS and NG5, where the ice is better resolved and the velocity field is rougher. The recommendation of $K = 4$ should be read as scoped to CORE2 until a high-resolution year has been run. This is the largest gap.
- **The divergence form over a long integration.** Since it showed no speed benefit we did not carry it through a one-year test, so the 2.1×10^{-11} per-step non-equivalence of section 6.1 has not been followed to its long-term consequences.
- **Core counts above 2304 on fArc.** The 2019 paper reaches 6912. The partition set we have for that mesh stops at 2304, and the separately partitioned copy in the shared archive does not run in this port (the ocean solver fails to converge immediately, at every time step tried, which is a configuration problem rather than a stability one). Our measured ice share reaches 13% at 2304 cores where the paper interpolates 25–30%; the two candidate reasons — this port’s own communication optimisations, and mEVP here versus standard EVP there — are untested.
- **The 13% GPU cost of the divergence form** (section 6.2) is measured but not explained. It is not arithmetic time and not the masked node kernel; we have no mechanism for it and are not offering one.

All configurations are switchable at run time in the port; the underlying runs, job scripts and the full internal write-up can be shared on request.